AQUAFAB INDUSTRIES

Swim Performance (Ucrit) Testing

The stepped-velocity test, with optional tracking and cost of transport.

For Aquafab Workbench 0.2.1

Edition of 29 October 2026

Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Swim performance (U_{crit}) testing

The critical swimming speed test of Brett (1964): water speed rises in fixed steps at fixed intervals until the animal can no longer hold station. You press **Fish Fatigued** at that moment, and U_{crit} is interpolated:

> $U_{crit} = U + (T / T_i) \times U_i$

U is the last speed held for a full interval, U_i the step size, T_i the interval and T the time at the final speed.

You need: a machine with a variable-speed pump (the **Swim Performance** tab appears only then), and a license to start a test on connected hardware. A camera is optional.

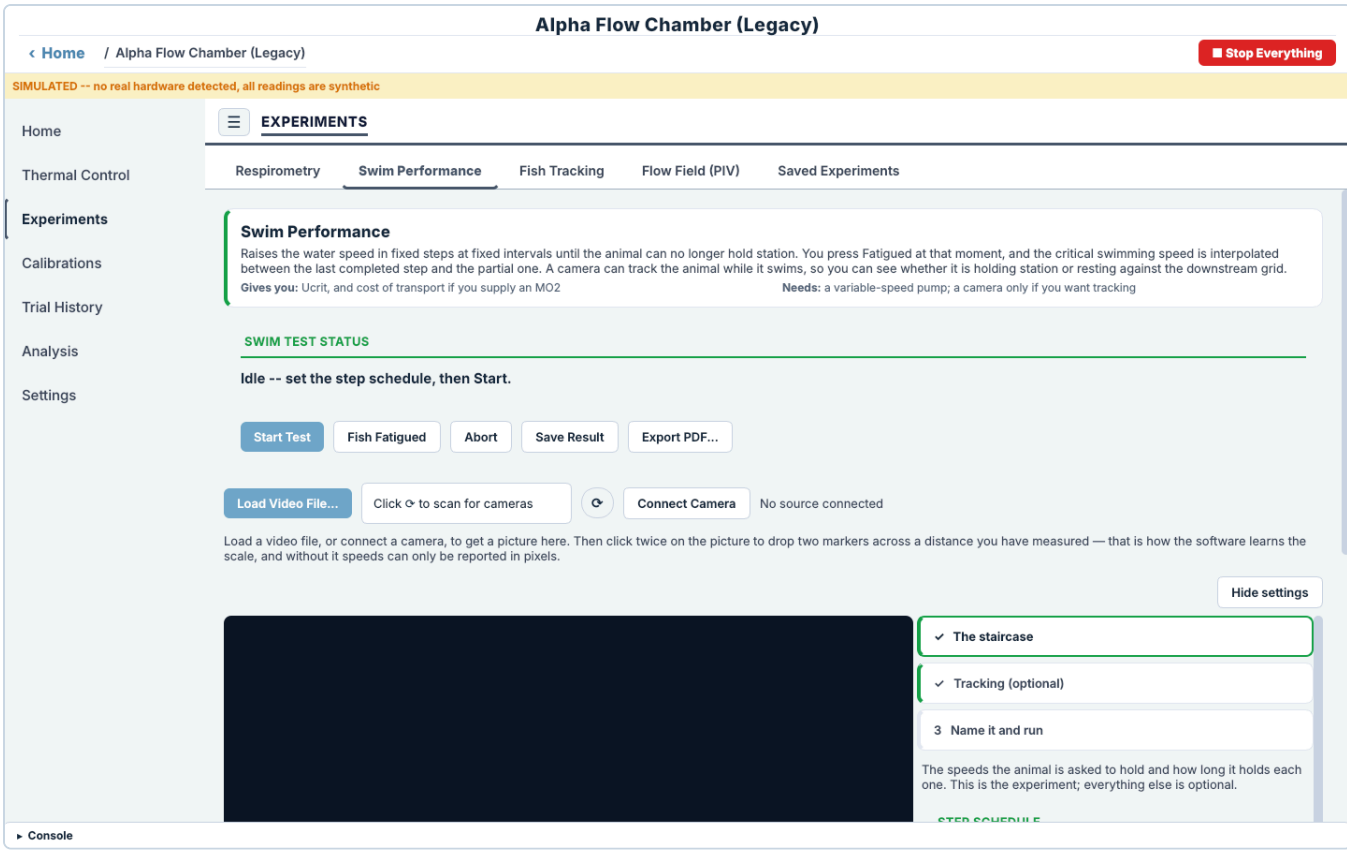
> Screenshots are from a simulated machine. Numbers in them are not measurements.

1.1 Before the day: calibrate velocity

The motor is commanded in per cent of output. To report U_{crit} in cm/s, the software needs to know how many cm/s one per cent produces. Measure it with the **Chamber Velocity (PIV)** calibration (*PIV Flow Measurement and Chamber Velocity Calibration*, Part 3). Without it, U_{crit} is reported in % output, which cannot be compared between machines.

1.2 Set up

Experiments → Swim Performance.



Swim Performance.

1.2.1 The staircase

Field	What to enter
Motor	The pump that drives the working section.
Start output (%)	The first speed.
Step size (%)	How much each step adds.
Interval T_i (s)	How long each speed is held (default 1200 s; published protocols range from minutes to an hour).
Maximum output (%)	The staircase never goes above this.
cm/s per % output	From the velocity calibration. 0 keeps results in %.
Fish length (cm)	For U_{crit} in body lengths per second. 0 skips it.
MO_2 at speed (mg/kg/h)	Only if you measure oxygen alongside: gives cost of transport at each step and U_{opt}, the cheapest speed. 0 skips it.
Simulation speed (x)	Simulated machines only: rehearse a long test in minutes.

1.2.2 Tracking (optional)

A track of the animal is the evidence for the moment you called fatigue.

1. **Connect Camera** above the picture.
2. With the tunnel **empty**, press **Capture background**.
3. Set the scale: click two points a known distance apart, type the distance, **Apply scale from markers**.
4. Tick **Track the animal during the test**.

The detection settings are explained in the *Fish Tracking* guide.

1.2.3 Name it and run

Name, **Treatment** and **Notes**. **Save experiment** keeps the staircase for the next animal.

1.3 Run

1. Put the animal in and let it settle at a low speed, as your protocol requires.
2. Press **Start Test**. The status shows the current speed and the time left in the step.
3. Watch the **animal**, not the screen. Decide your fatigue criterion before the test (typically: resting against the downstream grid and not resuming after gentle encouragement) and apply it the same way every time.
4. The moment it fatigues, press **Fish Fatigued**, or the **F** key. The motor stops.

Abort stops the test and the motor without a result. If the staircase reaches **Maximum output**, it holds there and keeps timing until you press Fish Fatigued or Abort.

1.4 The result

The **Result** card shows U_{crit} in cm/s (if calibrated), in body lengths per second (if a length was given) and in % output; the completed steps and the time at the final speed; and, with an MO_2, the cost of transport at each step and U_{opt}.

Press **Save Result** to keep it in Trial History. **It is not saved until you do. Export PDF…** prints the protocol, result and speed chart (needs a license).

1.5 Good practice

- A large animal in a small working section swims in faster water than the nominal speed. The Fish Tracking page has a solid-blocking correction (Bell & Terhune 1970); verify its coefficients before publishing.
- Keep the interval and step size the same within a study. U_{crit} depends on both.